Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. We trained each model for 20 epochs with a batch size of 64.

2 Results

Figure 1 shows the accuracy by dataset.

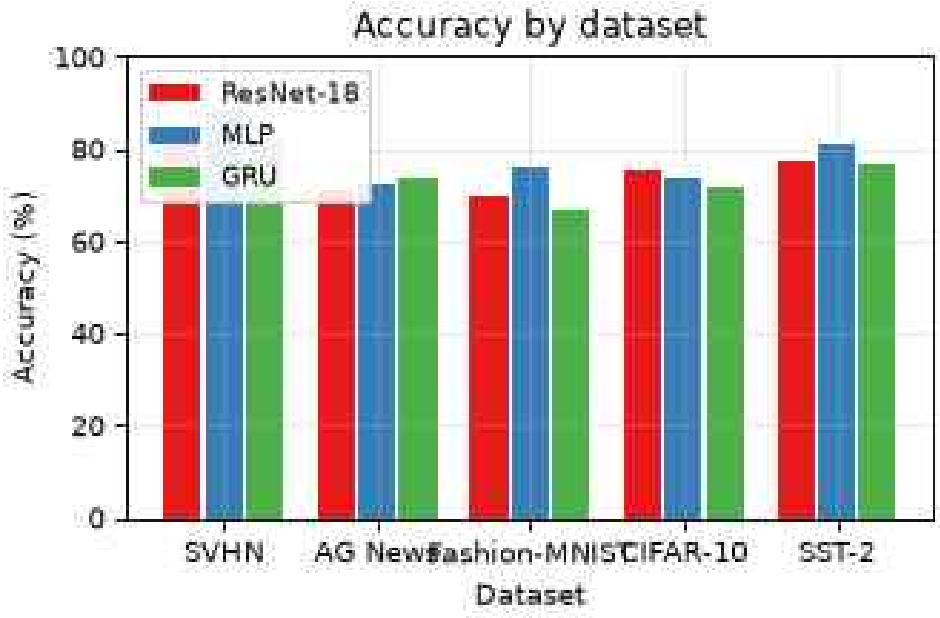


Figure 1: Accuracy by dataset for the models in this experiment.

On CIFAR-10, ResNet-18 obtains a accuracy of 75%. On CIFAR-10, ResNet-18 is 53.8 points better than GRU. MLP outperforms ResNet-18 by 1.3 percentage points on AG News. MLP reaches 76.3% accuracy on Fashion-MNIST (Figure 1).

Figure 2 shows the validation accuracy during training.

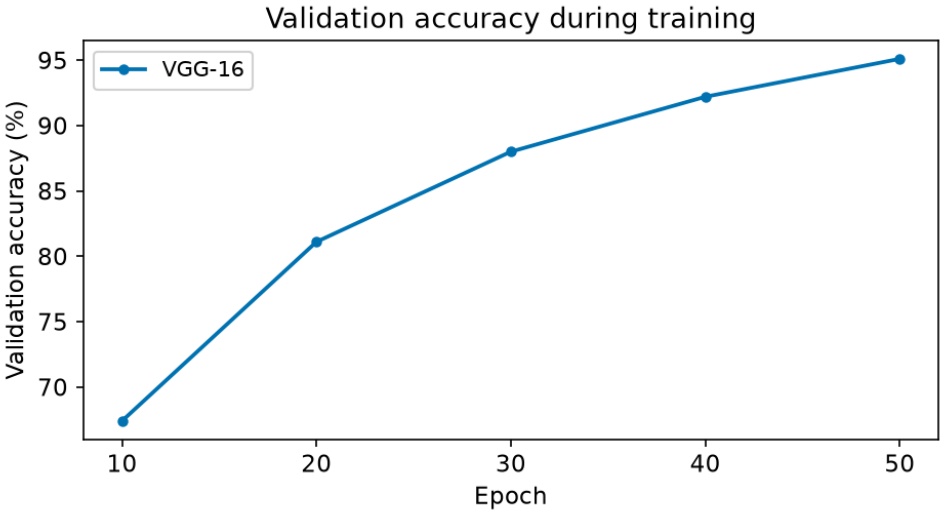


Figure 2: Validation accuracy during training for the models in this experiment.

From epoch 40 to epoch 50, the validation accuracy of VGG-16 improves by a relative 3.1%. From epoch 30 to epoch 40, the validation accuracy of VGG-16 improves by a relative 5%. VGG-16 reaches about 100% validation accuracy at epoch 50.

Figure 3 shows the accuracy of Baseline by dataset.

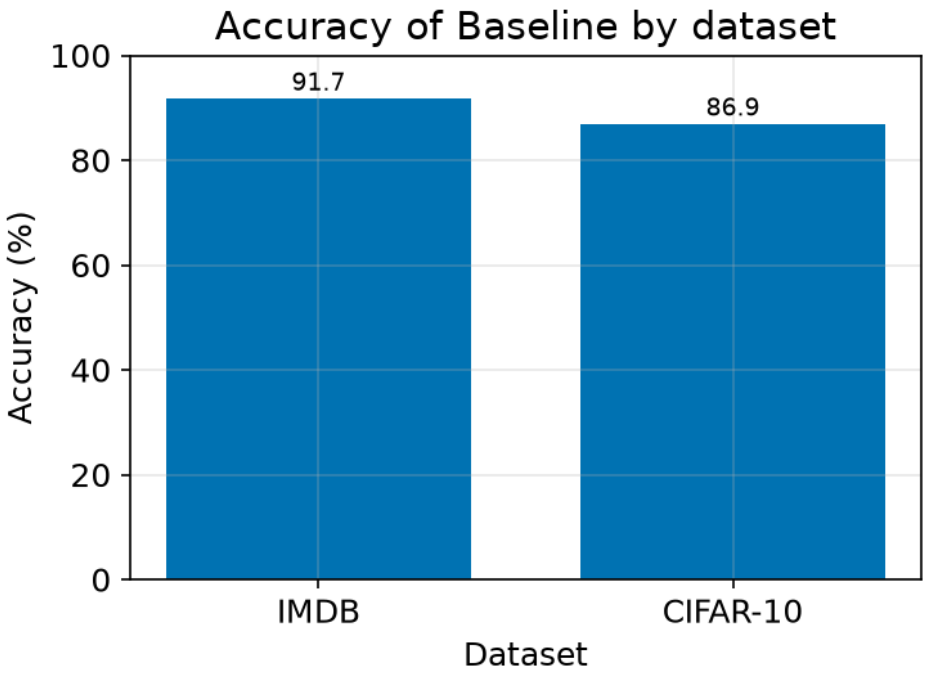


Figure 3: Accuracy of Baseline by dataset for the models in this experiment.

Baseline reaches 86.9% accuracy on IMDB. On IMDB, Baseline obtains a accuracy of 92%.

3 Discussion

We note that the training was stable for all models.